

SOUTHEAST IOWA, IA, USA

WMO: 725420

Lat: 40.783N

Lon: 91.125W

Elev: 211

StdP: 98.82

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14931

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-16.5	-23.7	0.5	-18.8	-21.2	0.6	-16.0	11.1	-4.7	10.1	-4.3	3.6	310	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	33.5	25.1	32.2	24.4	30.9	23.6	26.6	31.7	25.6	30.6	24.7	29.3	4.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.7	29.9	24.1	19.4	29.0	23.1	18.3	27.8	84.0	31.6	79.7	30.5	75.9	29.3	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.7	8.6	7.8	-23.5	35.7	3.7	1.8	-26.2	36.9	-28.4	38.0	-30.5	39.0	-33.2	40.3		
			-23.9	28.1	3.6	0.8	-26.5	28.7	-28.6	29.2	-30.6	29.7	-33.2	30.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-4.2	-1.6	4.9	11.6	17.5	22.6	24.4	23.5	19.7	12.8	5.4	-0.9	(6)
	DBStd	11.28	7.01	6.85	6.37	5.37	4.70	3.46	3.08	2.96	4.42	5.31	5.77	6.22	(7)
	HDD10.0	1533	441	329	183	43	3	0	0	0	1	32	158	342	(8)
	HDD18.3	3175	699	559	418	209	74	6	1	2	37	185	387	598	(9)
	CDD10.0	2030	0	3	25	92	234	378	447	417	291	119	22	2	(10)
	CDD18.3	632	0	0	1	8	48	135	189	161	76	13	1	0	(11)
	CDH23.3	5682	0	0	10	82	397	1201	1853	1365	667	105	2	0	(12)
	CDH26.7	1999	0	0	0	13	107	424	727	477	228	23	0	0	(13)
(14) Wind	WSAvg	3.8	4.2	4.2	4.5	4.6	3.9	3.3	2.8	2.7	3.1	3.7	4.2	4.0	(14)
(15) Precipitation	PrecAvg	879	41	29	73	97	92	98	95	90	105	67	50	43	(15)
	PrecMax	1273	100	52	149	189	177	228	184	206	296	184	90	126	(16)
	PrecMin	657	6	4	20	17	27	24	30	15	0	2	5	2	(17)
	PrecStd	165	27	15	34	51	40	51	41	54	78	43	27	28	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.4	18.2	25.2	28.6	32.1	33.9	36.4	34.6	33.6	29.6	22.6	17.1	(19)
		MCWB	10.3	12.8	17.1	18.5	22.7	24.8	26.4	25.4	23.4	21.2	15.3	14.5	(20)
	2%	DB	10.3	14.0	21.2	25.9	29.4	32.5	33.9	32.7	31.3	26.2	19.0	13.4	(21)
		MCWB	7.4	10.0	14.9	17.0	21.2	24.2	25.9	25.0	22.5	18.6	13.7	10.5	(22)
	5%	DB	7.4	10.9	17.7	23.2	27.5	31.0	32.2	31.1	29.3	23.6	16.7	10.1	(23)
		MCWB	4.8	7.7	12.6	15.6	19.8	23.4	25.2	24.2	21.7	17.3	12.3	7.8	(24)
	10%	DB	4.5	7.6	14.5	20.9	25.5	29.1	30.6	29.3	27.4	21.2	14.1	7.3	(25)
		MCWB	2.6	5.0	10.0	14.2	18.7	22.5	24.1	23.2	20.6	16.3	10.4	5.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	14.2	18.1	20.1	24.1	26.6	28.1	27.4	25.1	22.3	17.0	15.1	(27)
		MCDB	13.3	16.9	23.9	26.3	29.8	32.1	33.4	32.8	30.9	27.8	20.5	17.3	(28)
	2%	WB	7.6	10.5	15.8	18.2	22.4	25.4	26.9	26.1	23.9	20.4	14.9	11.0	(29)
		MCDB	10.2	13.7	19.9	23.6	27.8	31.0	32.0	31.0	29.2	24.1	17.8	12.4	(30)
	5%	WB	4.9	7.7	13.5	16.8	21.2	24.5	25.9	25.1	22.9	18.7	13.0	7.8	(31)
		MCDB	7.3	10.4	17.2	21.9	26.0	29.4	30.8	29.7	27.2	22.0	15.9	9.8	(32)
	10%	WB	2.8	5.1	10.4	15.1	19.9	23.5	25.0	24.1	21.8	16.9	10.8	5.2	(33)
		MCDB	4.5	7.7	14.1	19.9	24.0	28.0	29.6	28.1	26.0	20.7	13.8	7.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.8	10.1	11.1	10.6	10.3	10.1	10.2	11.6	10.8	9.4	8.0	(35)
		MCDBR	11.6	12.9	13.7	14.4	12.4	11.6	11.2	11.6	13.1	13.4	12.4	10.8	(36)
	5% WB	MCWBR	9.2	9.2	8.6	8.1	6.4	5.6	5.2	5.5	5.8	7.2	8.5	8.8	(37)
		MCWBR	10.8	11.6	12.5	12.5	11.0	10.4	10.2	10.2	10.9	10.6	11.2	10.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.290	0.310	0.340	0.388	0.422	0.426	0.443	0.431	0.395	0.350	0.303	0.290	(40)
		taud	2.453	2.395	2.400	2.267	2.234	2.281	2.205	2.254	2.310	2.438	2.511	2.465	(41)
	Ebn at Noon		875	904	912	884	857	853	833	833	840	846	854	844	(42)
		Edh at Noon	77	95	106	130	138	132	141	131	116	91	73	71	(43)
(44) All-Sky Solar Radiation	RadAvg		1.83	2.79	3.72	4.75	5.55	6.22	6.22	5.46	4.68	3.13	2.10	1.55	(44)
	RadStd		0.21	0.24	0.27	0.38	0.40	0.57	0.39	0.31	0.35	0.35	0.17	0.19	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page